Multivariable Calculus and Differential Equation Assignment

Title: Evaluating Triple Integrals in Spherical Coordinates & 3D Octave Visualization
A computational and mathematical analysis assignment focused on evaluating the total pixel intensity over a spherical object (R = 4) using the intensity function f(x,y,z) = x^2 + y^2 + z^2 + 1. The task bridges multivariable calculus with programming by converting Cartesian coordinates into spherical coordinates and utilizing GNU Octave to compute exact/numerical triple integrals while generating 3D vector-field visualizations of the region of integration.

Skills and Knowledge Shown:
• Advanced multivariable calculus (triple integrals, spherical coordinate transformations, Jacobian determinants).
• Scientific programming and symbolic math toolbox utilization in GNU Octave/MATLAB.
• 3D data visualization skills (meshgrid, quiver3, and surface mesh rendering).
• Analytical problem-solving applied to physical field simulations.

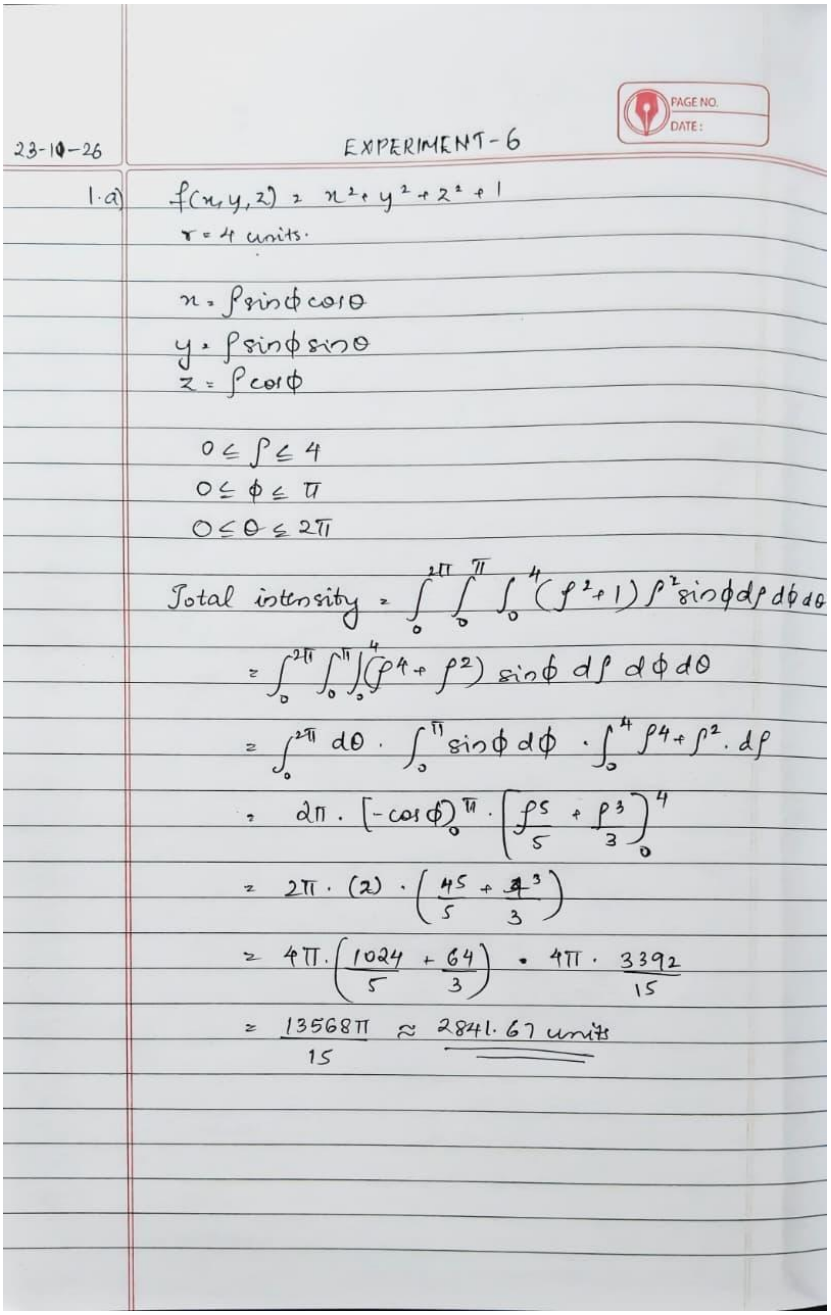
Experiment-6

Evaluating Triple Integrals

Aim:

To determine the total pixel intensity over a spherical object of radius 4 units by converting the given intensity function from Cartesian coordinates to spherical coordinates and to plot the region of integration using GNU Octave.

Mathematical Procedure:



23-10-26 EXPERIMENT-6

1.a) $f(x,y,z) = x^2+y^2+z^2+1$

$r = 4$ units.

$x = \rho \sin\phi \cos\theta$

$y = \rho \sin\phi \sin\theta$

$z = \rho \cos\phi$

$0 \le \rho \le 4$

$0 \le \phi \le \pi$

$0 \le \theta \le 2\pi$

Total intensity $= \int_0^{2\pi}\int_0^{\pi}\int_0^{4} (\rho^2+1)\rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

$= \int_0^{2\pi}\int_0^{\pi}\int_0^{4} (\rho^4+\rho^2)\sin\phi \, d\rho \, d\phi \, d\theta$

$= \int_0^{2\pi} d\theta \cdot \int_0^{\pi} \sin\phi \, d\phi \cdot \int_0^{4} \rho^4+\rho^2 \, d\rho$

$= 2\pi \cdot [-\cos\phi]_0^{\pi} \cdot \left[\frac{\rho^5}{5}+\frac{\rho^3}{3}\right]_0^{4}$

$= 2\pi \cdot (2) \cdot \left(\frac{4^5}{5}+\frac{4^3}{3}\right)$

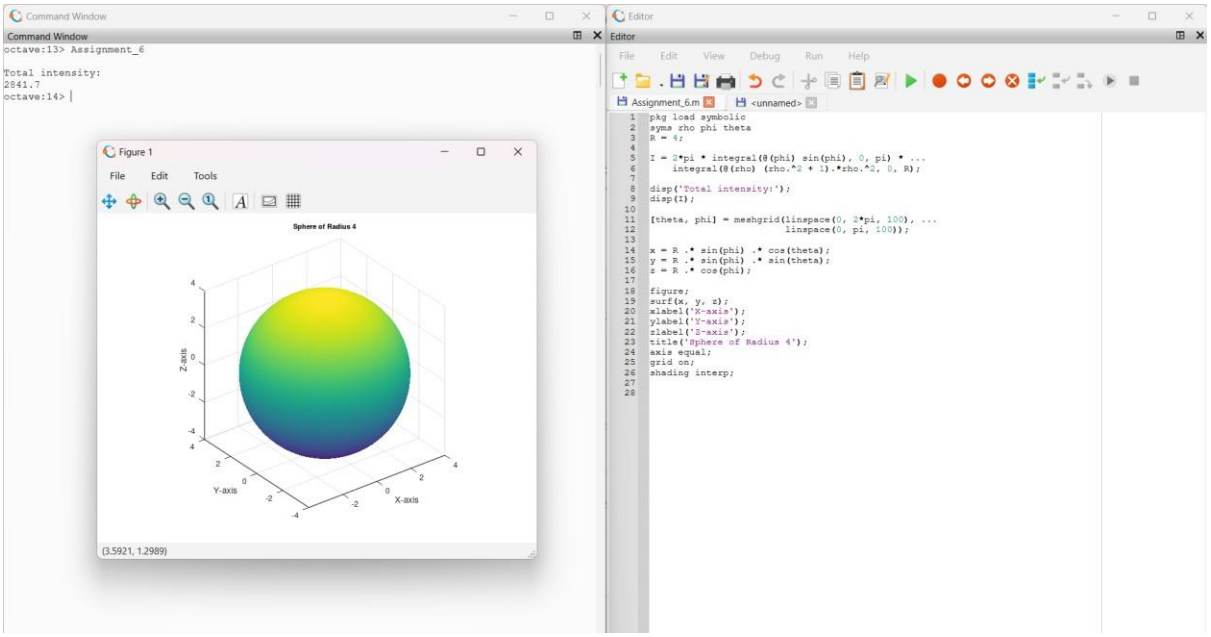
$= 4\pi \cdot \left(\frac{1024}{5}+\frac{64}{3}\right) = 4\pi \cdot \frac{3392}{15}$

$= \frac{13568\pi}{15} \approx 2841.67$ units

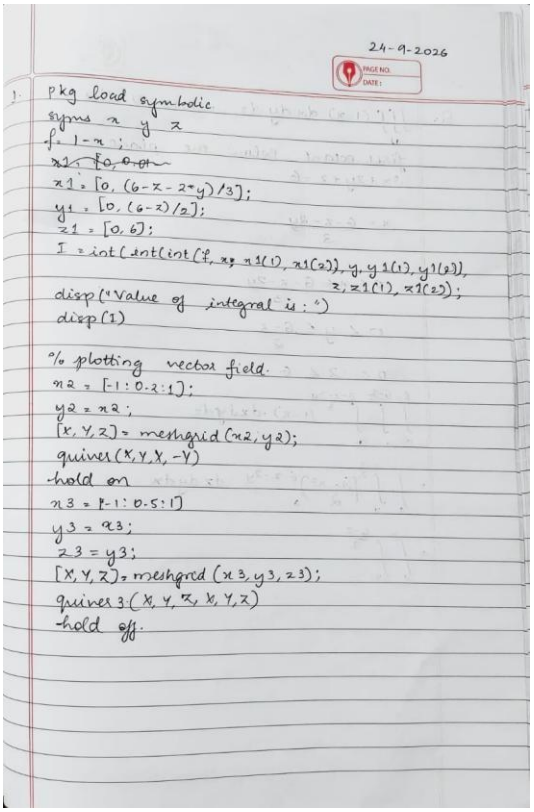
OCTAVE CODE:

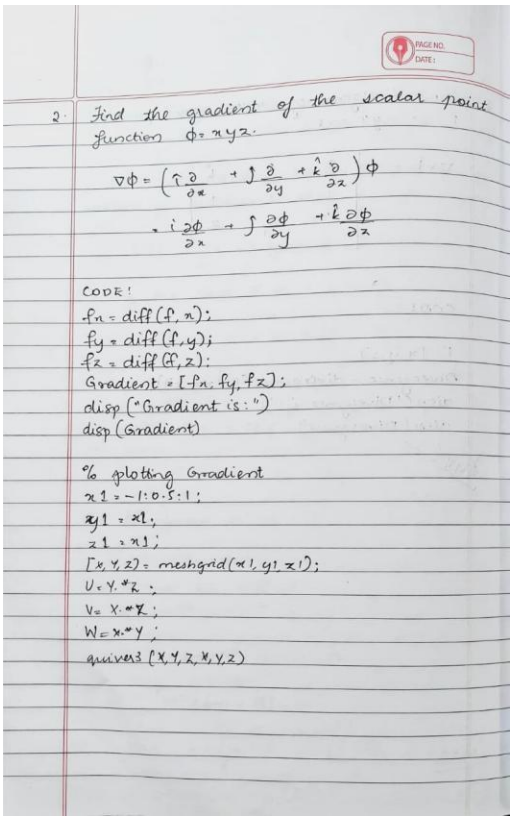
```
pkg load symbolic
syms rho phi theta
R = 4;
I = 2*pi * integral(@(phi) sin(phi), 0, pi) * ...
   integral(@(rho) (rho.^2 + 1).*rho.^2, 0, R);
disp('Total intensity:');
disp(I);
[theta, phi] = meshgrid(linspace(0, 2*pi, 100), ...
                        linspace(0, pi, 100));
x = R .* sin(phi) .* cos(theta);
y = R .* sin(phi) .* sin(theta);
z = R .* cos(phi);
figure;
surf(x, y, z);
xlabel('X-axis');
ylabel('Y-axis');
zlabel('Z-axis');
title('Sphere of Radius 4');
axis equal;
grid on;
shading interp;
```

Output:



Lab Signed Copy:





2. Find the gradient of the scalar point function $\phi = xyz$.

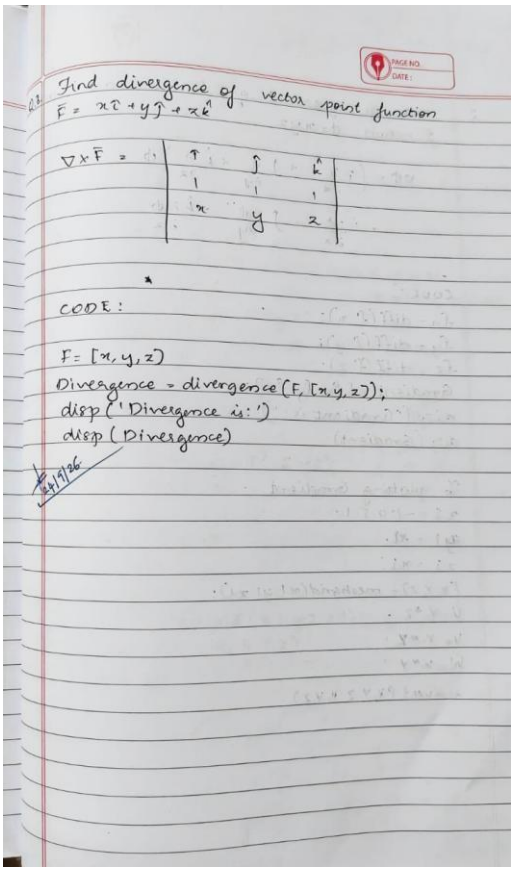
$$\nabla\phi = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\phi$$

$$= \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$$

CODE:

```
fx = diff(f, x);
fy = diff(f,y);
fz = diff(f,z);
Gradient = [fx, fy, fz];
disp("Gradient is:")
disp(Gradient)

% plotting Gradient
x1 = -1:0.5:1;
y1 = x1;
z1 = x1;
[x, y, z] = meshgrid(x1, y1, z1);
U = y.*z;
V = x.*z;
W = x.*y;
quiver3(x,y,z,u,v,w)
```



Find divergence of vector point function $\bar{F} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\nabla \times \bar{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

CODE:

```
F = [x, y, z]
Divergence = divergence(F, [x, y, z]);
disp('Divergence is:')
disp(Divergence)
```

24/9/26

(Three questions were done on 24th in lab, out of which only 1st question matched this experiment to an extent. I uploaded all here so sign could be included. The rest 2 questions are done in the 7th experiment.)